



assertEquals() vs. assertEquals() in JUnit

Last Updated : 23 Jul, 2025

JUnit is a popular testing framework widely utilized for developing and executing tests in Java applications. When writing unit tests, it is essential to verify that the outcomes of the code behave as expected. JUnit provides various assertion methods to help verify these outcomes, with two of the most common being `assertEquals()` and `assertEquals()`.

In unit testing, assertions are used to check whether the actual result of a piece of code matches the expected result. JUnit provides several assertion methods, and two commonly used ones are `assertEquals()` and `assertEquals()`. Although they may seem similar, their purposes and the way they compare objects are different. In this article, we will learn the difference between `assertEquals()` and `assertEquals()` in JUnit.

assertEquals() in JUnit

The `assertEquals()` method in JUnit is used to compare the values of two objects or primitive types. It checks whether the provided values are logically equal. This method is commonly used to verify that the content of an object matches the expectation.

Syntax:

```
assertEquals(expected, actual);
```

Example:

```
@Test
public void testAssertEquals() {
    String actual = new String("JUnit");
```

```
String expected = new String("JUnit");

// This will pass because both strings have the same value.
assertEquals(expected, actual);
}
```

In this example, even though the expected and actual refer to different objects (i.e., different memory locations), `assertEquals()` will pass because the value of the two strings is the same.

When to Use assertEquals()

- When you want to compare the contents of two objects.
- When you want to compare primitive data types.
- When equality of the objects is more important than identity.

assertSame() in JUnit

The `assertSame()` method checks whether two objects refer to the same instance in memory. It tests whether the two objects are identical in terms of their memory address, rather than comparing their values.

Syntax:

```
assertSame(expected, actual);
```

Example:

```
@Test
public void testAssertSame() {
    String actual = "JUnit";
    String expected = actual;

    // This will pass because both refer to the same memory
    location.
    assertSame(expected, actual);
}
```

In this example, both expected and actual refer to the same memory location; therefore, assertEquals() will pass.

When to Use assertEquals()

- When you need to verify that two references point to the same object in memory.
- When checking object identity is important.
- When you need to ensure that no new object instance has been created unnecessarily.

Difference Between assertEquals() and assertEquals()

Here is a detailed comparison of assertEquals() and assertEquals() in JUnit:

Feature	assertEquals()	assertEquals()
Purpose	Compares the values of two objects or primitives.	Compares if two references point to the same object in memory.
Comparison Type	Logical equality (via equals() method).	Checks if two references point to the same object.
Use Case	Used when the content or value of the two objects needs to be checked.	Used when checking if two references are pointing to the same object.
Primitive Type Support	Supports comparison of both objects and primitive types.	Only supports object reference comparison (no primitives).

Feature	assertEquals()	assertSame()
Object Comparison	Compares object values (using the equals() method if overridden).	Compares if two references are pointing to the same object instance.
Example (String)	assertEquals(new String("JUnit"), new String("JUnit")) will pass as both values are equal.	assertSame(new String("JUnit"), new String("JUnit")) will fail because they are different instances.
Example (Primitive)	assertEquals(10, 10) will pass as the primitive values are the same.	assertSame(10, 10) will throw a compilation error since primitives can't be compared by reference.
Object Identity	Does not verify if the objects are identical in memory.	Verifies if the objects are the exact same reference.
Failure Case (Object)	Fails when equals() method returns false.	Verifies if the objects are the exact same reference.
Null Comparison	Passes if both values are null.	Passes only if both references are null or point to the same object.

Example 1: Testing Object Identity with assertEquals()

```
@Test
public void testObjectIdentity() {
```

```
Integer num1 = Integer.valueOf(5);
Integer num2 = Integer.valueOf(5);

// This will pass as both refer to the same object in memory
// due to integer caching.
assertEquals(num1, num2);
}
```

Example 2: Testing Value Equality with assertEquals()

```
@Test
public void testValueEquality() {
    Integer num1 = new Integer(5);
    Integer num2 = new Integer(5);

    // This will pass as their values are the same.
    assertEquals(num1, num2);

    // This will fail as they are different objects.
    assertEquals(num1, num2);
}
```

Conclusion

In JUnit, both `assertEquals()` and `assertSame()` are useful methods for different purposes. While `assertEquals()` focuses on comparing the values of objects, `assertSame()` checks whether the two references point to the same object. Understanding when to use each assertion method is crucial for writing effective and reliable unit tests.

[Comment](#)**M** mahes... [+ Follow](#)

0

Article Tags :

[Advance Java](#)[JUnit](#)

Explore

Java Enterprise Edition

Multithreading

Concurrency

JDBC (Java Database Connectivity)

Java Frameworks

JUnit



Corporate & Communications Address:

A-143, 7th Floor, Sovereign Corporate Tower, Sector- 136, Noida, Uttar Pradesh (201305)

Registered Address:

K 061, Tower K, Gulshan Vivante Apartment, Sector 137, Noida, Gautam Buddh Nagar, Uttar Pradesh, 201305



Company

[About Us](#)
[Legal](#)
[Privacy Policy](#)
[Contact Us](#)
[Advertise with us](#)
[GFG Corporate Solution](#)

Explore

[POTD](#)
[Job-A-Thon](#)
[Blogs](#)
[Nation Skill Up](#)

Campus Training Program

Tutorials

Programming Languages

DSA

Web Technology

AI, ML & Data Science

DevOps

CS Core Subjects

Interview Preparation

Software and Tools

Videos

DSA

Python

Java

C++

Web Development

Data Science

CS Subjects

Courses

IBM Certification

DSA and Placements

Web Development

Programming Languages

DevOps & Cloud

GATE

Trending Technologies

Preparation Corner

Interview Corner

Aptitude

Puzzles

GfG 160

System Design

@GeeksforGeeks, Sanchhaya Education Private Limited, All rights reserved